PV102 Day 6 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 22 | Morning Quiz - Answer Key
Power electronics set system boundaries. | Tuesday, August 11, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. An inverter primarily converts:

- A. DC to AC
- B. AC to sunlight
- C. wood to wire
- D. shade to volts

Answer: A

Rationale/look-for: Inverters convert DC energy to AC output for AC loads.

2. A charge controller protects and manages:

- A. Battery charging from PV
- B. Roof pitch
- C. Clipboards
- D. Nameplate stickers only

Answer: A

Rationale/look-for: Controllers regulate charging within battery and PV limits.

3. Inverter sizing must consider:

- A. Continuous load and surge
- B. Marker color only
- C. Roof direction only
- D. Lunch menu

Answer: A

Rationale/look-for: Continuous and starting/surge requirements both matter.

4. An overcurrent device is used to:

- A. Open a circuit during excessive current
- B. Store solar energy
- C. Increase sunshine
- D. Convert AC to shade

Answer: A

Rationale/look-for: OCPDs limit damaging current when properly selected and installed.

5. A disconnect provides:

- A. A means to isolate equipment
- B. Free battery capacity
- C. More roof area
- D. A better compass

Answer: A

Rationale/look-for: Disconnects support isolation for service and emergency boundaries.

6. Why should source, storage, conversion, and load boundaries be shown separately?

_____

_____

Answer: Clarity/safety

Rationale/look-for: It clarifies what each component controls and where hazards/isolations exist.

7. Name one label that belongs on a training system handoff diagram.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept voltage, current, source, battery, disconnect, OCPD rating, load panel, do not energize, polarity, or owner/equipment ID.

8. Why can an inverter with enough continuous watts still fail a job?

_____

_____

Answer: Surge/output/voltage

Rationale/look-for: Surge, output voltage, waveform, DC input, or environmental limits may not match.

9. What is one thing students should not decide without manufacturer instructions?

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept torque, wiring sequence, fuse size, grounding detail, settings, charge profile.

10. Write the Day 6 rule in your own words.

_____

_____

Answer: Power electronics boundaries

Rationale/look-for: Look for controller/inverter as hard limits and conversion boundaries.